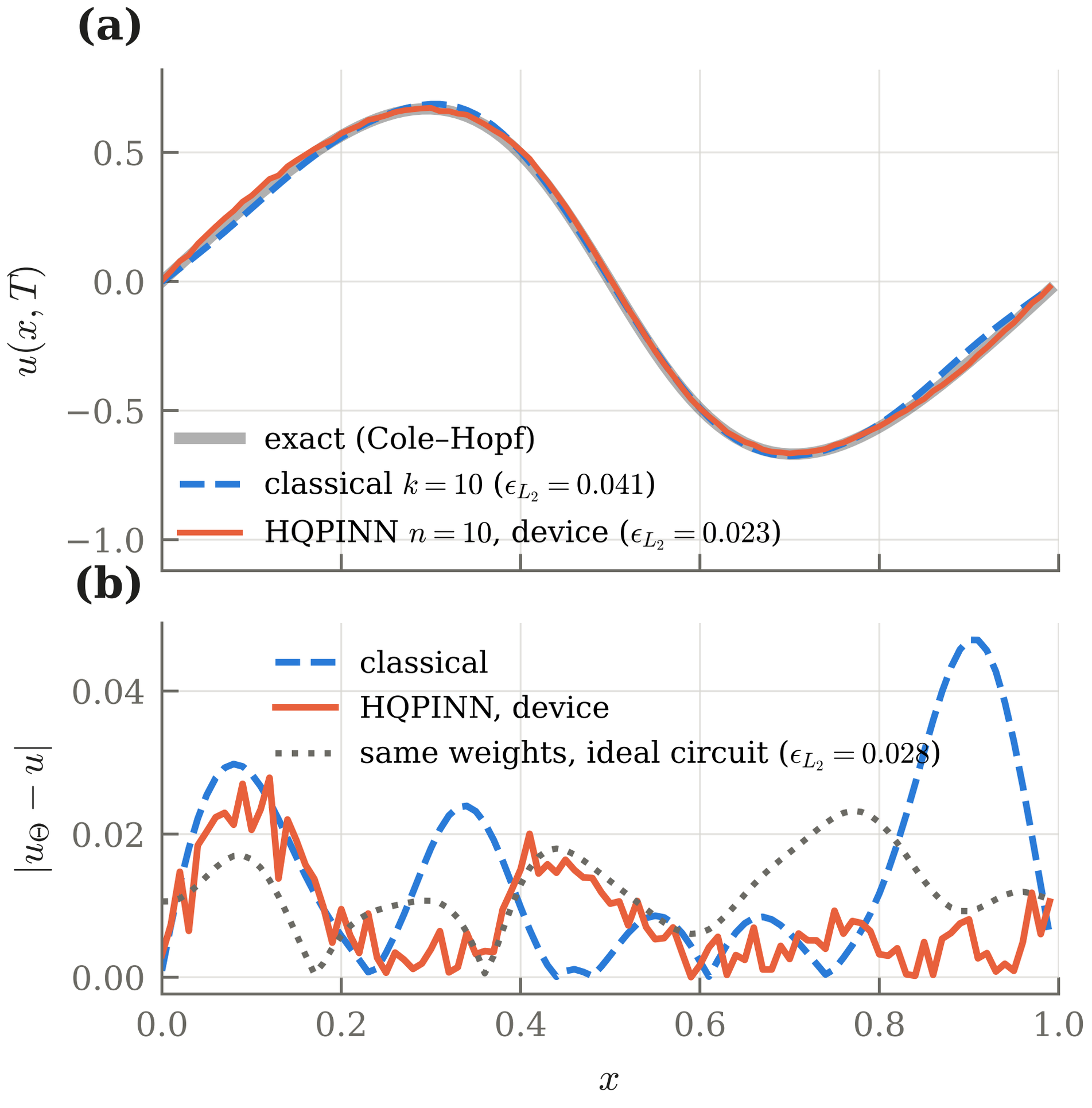
(a)
0.5
0.0
−0.5
−1.0
$u(x, T)$
exact (Cole–Hopf)
classical $k = 10$ ($\epsilon_{L_2} = 0.041$)
HQPINN $n = 10$, device ($\epsilon_{L_2} = 0.023$)
(b)
0.04
0.02
0.00
$|u_\Theta - u|$
classical
HQPINN, device
same weights, ideal circuit ($\epsilon_{L_2} = 0.028$)
0.0
0.2
0.4
0.6
0.8
1.0
x